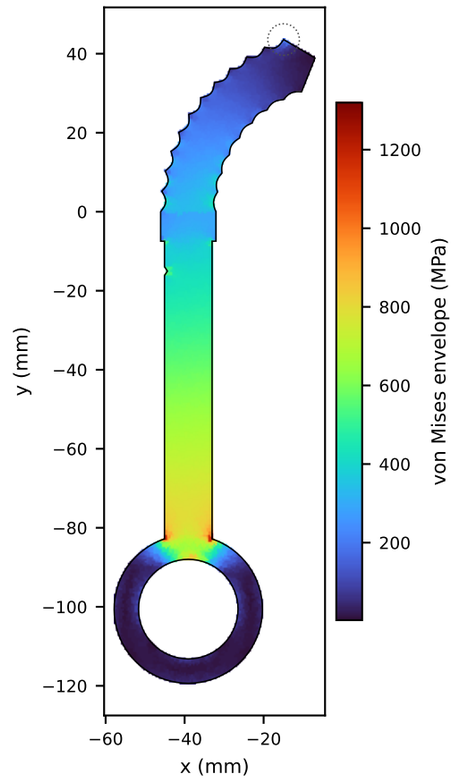
Wave 3 — fem_smoke (plate-FEM-in-loop smoke) | Fitness -7.248 N/g | Est. weight 44.8 g

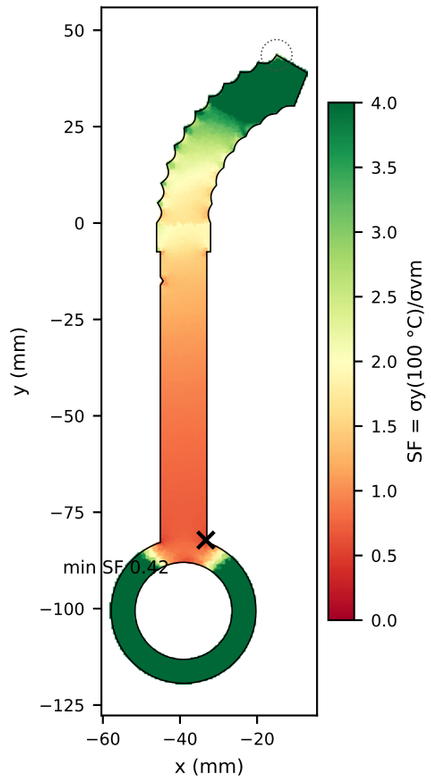
FEM-UNCALIBRATED: offline Kirchhoff plate (numerical DKT + CST, 6016 tris) — not yet calibrated to MC / repo FEM. Envelope of hunter-lateral 260 N, twist 1.26 N·m, tip-pry 260 N, ring-pull 1500 N.

FEM-UNCALIBRATED **QUARANTINED: blade < 3.375 in (all three)** **hard geometric gates: red-team Ok** **floors 180/260/1500 N on plate: FAIL**

(1) von Mises — worst load case per element



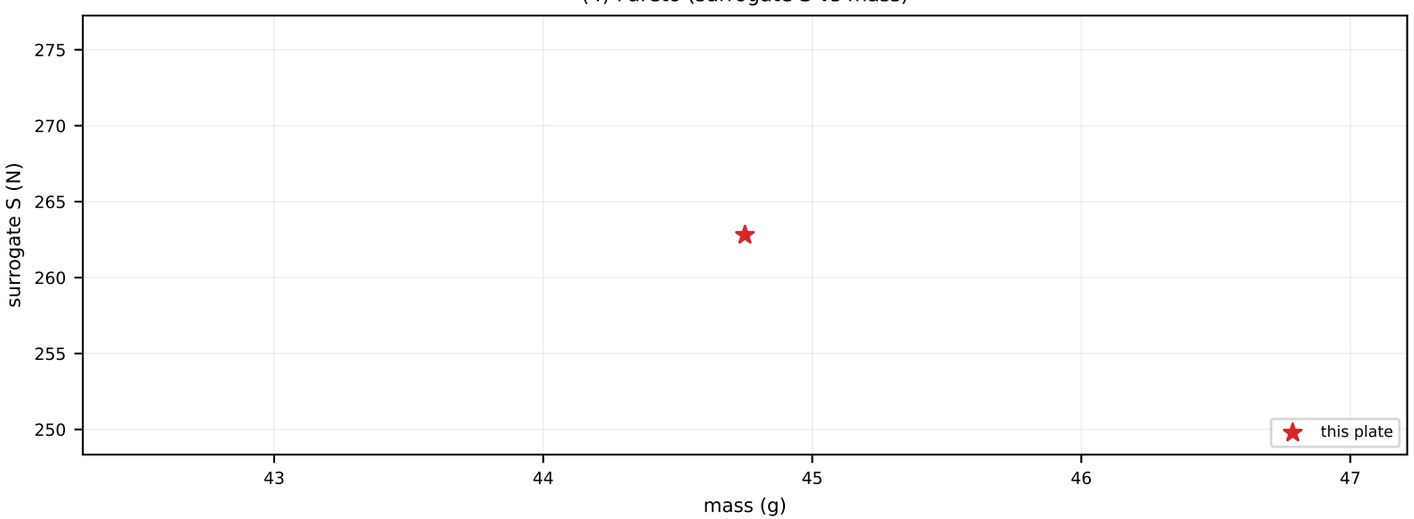
(2) Safety factor — worst case @100 °C



(3) Metrics

Metric	Value	Gate / note
Blade tip→heel	2.171 in (55.1 mm)	**≥3.375 in: FAIL**
Blade horizontal reach	1.539 in (39.1 mm)	**≥3.375 in: FAIL**
Blade edge arc (outer)	2.613 in (66.4 mm)	**≥3.375 in: FAIL**
Grip (ring-neck→guard)	75.3 mm	60–80 mm
Ring ID	25.19 mm	25.4±0.5 mm
Stock / material	4.4 mm / Ti-3Al-2.5V	red-team: Ok
Est. weight	44.75 g	geom check 44.76 g
Fitness	-7.248 N/g	robust plate N/g (FEM-in-loop)
Beam surrogate S	263 N	repo plate (fem.rs): 106 N
Plate FEM @ -10 °C	lat 127 N | ring 5649 N	SF hiker 0.70 / hunter 0.49 / ring 3.77
Plate FEM @ +20 °C	lat 122 N | ring 5453 N	SF hiker 0.68 / hunter 0.47 / ring 3.64
Plate FEM @ +100 °C	lat 110 N | ring 4929 N	SF hiker 0.61 / hunter 0.42 / ring 3.29
Twist / pry SF @100 °C	twist 7.97 | pry 1.29	twist 1.26 N·m, pry 260 N

(4) Pareto (surrogate S vs mass)



Floors on offline plate, worst of -10/20/100 °C: hiker FAIL · hunter FAIL · ring PASS
Temp: Ti yield linear derate (-0.12 %/°C from 20 °C); maps use 100 °C yield 560 MPa.
Serration knock-down on blade (h/h_eff=1.16, from history genes). Load patch r≤4 mm excluded from peaks.